

A Norm-Increasing 4-Isometry Without Wold-Type Decomposition

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Status. Candidate full counterexample, likely valid, to Shimorin's question as quoted by Chavan and Trivedi.

1 Question and answer

For $T \in \mathcal{B}(\mathcal{H})$ put

$$\Delta_{r,T} = \sum_{k=0}^r (-1)^k \binom{r}{k} T^{*k} T^k.$$

The source calls T norm-increasing when $\Delta_{1,T} \leq 0$, and r -concave when $(-1)^r \Delta_{r,T} \leq 0$. It recalls Shimorin's question: must every norm-increasing r -concave operator, $r \geq 3$, have Wold-type decomposition?

and formally introduced in [17] Eq. (1.6)].

Given a positive integer m and $T \in \mathcal{B}(\mathcal{H})$, set

$$\Delta_{m,T} := \sum_{k=0}^m (-1)^k \binom{m}{k} T^{*k} T^k.$$

An operator T is said to be an *isometry* (resp. *norm-increasing*) if $\Delta_{1,T} = 0$ (resp. $\Delta_{1,T} \leq 0$). For $m \geq 2$, an operator $T \in \mathcal{B}(\mathcal{H})$ is said to be an *m -isometry* (resp. *m -concave*, resp. *m -expansion*) if $\Delta_{m,T} = 0$ (resp. $(-1)^m \Delta_{m,T} \leq 0$, resp. $\Delta_{m,T} \leq 0$). The reader is referred to [1, 16, 17] for the basic properties of m -isometries and related classes.

In [17] p. 185], Shimorin asked whether it is true that the m -concave and norm-increasing operators have Wold-type decomposition for $m \geq 3$. Here we explain the thought process that could help in answering this question. In this regard, note that the construction in [4] Example 3.1] was capitalized on the idea of the weighted shift on a one-circuit directed graph (see [7] Section 3]). The one-dimensional space on which the weighted shift in question fails to be norm-increasing corresponds to the indicator function of an element of the circuit. Thus, to get a counterexample to Shimorin's question from weighted shifts on directed graphs, it is necessary to confine ourselves to directed graphs without circuits. This thinking can indeed be strongly based on [4] Theorem 4.3], which provide a necessary condition for the failure of the wandering subspace property for analytic norm-increasing m -concave

Source excerpt: arXiv:2501.01296, p. 2 (locally compiled from the arXiv source).

Theorem 1. *There is a bounded analytic norm-increasing 4-isometry S which does not have Wold-type decomposition. Consequently Shimorin's question has a negative answer. The same S is an r -isometry, hence r -concave, for every $r \geq 4$.*

2 Proof intuition

Chavan and Trivedi construct norm-increasing shifts without Wold decomposition on a rootless quasi-Brownian tree. Quadratic moments make the third defect vanish off the tree's spine. Their constant-coefficient example has a nonzero third defect on the spine and therefore is only a 3-expansion, with the wrong sign for Shimorin's problem.

The extra observation is to ask for the *next* defect to vanish. Since

$$\Delta_{4,S} = \Delta_{3,S} - S^* \Delta_{3,S} S,$$

the spine values of the third defect need only be transported by the squared spine weights. Changing one linear coefficient, at level $m = 2$, makes those values exactly -2 below that level and $-2/m$ above it. This is precisely the transport law. The coefficient change leaves intact the source's convergent Cauchy-dual series, which is the obstruction to Wold decomposition.

3 The weighted shift

Let $V = \mathbb{N} \times \mathbb{Z}$. Give V the rootless directed-tree structure

$$\text{Chi}(0, m) = \{(0, m-1), (1, m)\}, \quad \text{Chi}(n, m) = \{(n+1, m)\} \quad (n \geq 1).$$

Set

$$a_m = \begin{cases} 1, & m = 2, \\ 2, & m \neq 2, \end{cases} \quad p_m(j) = 1 + a_m j + j^2 \quad (j \in \mathbb{N}).$$

The weight indexed by a child (n, m) is

$$\lambda_{(n,m)} = \begin{cases} \sqrt{p_m(n-1)/p_m(n-2)}, & n \geq 2, \\ \sqrt{m/(m+1)}, & n = 0, m \geq 1, \\ 1/\sqrt{m}, & n = 1, m \geq 1, \\ 1, & n \in \{0, 1\}, m \leq 0. \end{cases}$$

Define $Se_v = \sum_{w \in \text{Chi}(v)} \lambda_w e_w$ on $\ell^2(V)$.

These are the weights of Proposition 5.1 of the source, with quadratic coefficients bounded above and with quadratic coefficient uniformly equal to one. For completeness, the one-step norms are

$$\|Se_{(0,m)}\|^2 = \begin{cases} 1, & m \geq 2, \\ 2, & m \leq 1, \end{cases} \quad \|Se_{(n,m)}\|^2 = \frac{p_m(n)}{p_m(n-1)} \geq 1 \quad (n \geq 1).$$

Moreover $p_m(n)/p_m(n-1) \leq 4$. Thus S is bounded and $S^*S \geq I$, so it is norm-increasing and left-invertible.

4 Exact fourth-defect computation

Descendant sets at a fixed depth are disjoint for distinct vertices. Hence $S^{*k}S^k$, and therefore every $\Delta_{r,S}$, is diagonal in the canonical basis.

If $n \geq 1$, direct telescoping gives

$$\|S^k e_{(n,m)}\|^2 = \frac{p_m(n+k-1)}{p_m(n-1)}.$$

This is quadratic in k , so both its third and fourth finite differences vanish. Thus $\Delta_{3,S}$ and $\Delta_{4,S}$ vanish off the spine.

Let

$$d_m = \langle \Delta_{3,S} e_{(0,m)}, e_{(0,m)} \rangle.$$

Substitution of $b_m = 1$ and the above a_m in the exact spine calculation from the source gives

$$d_m = \begin{cases} -2, & m \leq 1, \\ -2/m, & m \geq 2. \end{cases}$$

The branch child $(1, m)$ has zero third-defect entry. The squared weight of the spine child $(0, m-1)$ is

$$\lambda_{(0,m-1)}^2 = \begin{cases} 1, & m \leq 1, \\ (m-1)/m, & m \geq 2. \end{cases}$$

Using $\Delta_{4,S} = \Delta_{3,S} - S^* \Delta_{3,S} S$, its spine entry is therefore

$$d_m - \lambda_{(0,m-1)}^2 d_{m-1} = 0$$

for every $m \in \mathbb{Z}$: for $m \leq 1$ both d 's equal -2 ; for $m = 2$ the product is $\frac{1}{2}(-2) = -1 = d_2$; and for $m \geq 3$ it is $\frac{m-1}{m} \frac{-2}{m-1} = -2/m = d_m$. Every diagonal entry of $\Delta_{4,S}$ is zero, hence $\Delta_{4,S} = 0$.

Finally, $\Delta_{r+1,S} = \Delta_{r,S} - S^* \Delta_{r,S} S$, so $\Delta_{r,S} = 0$ for all $r \geq 4$.

5 Analyticity and failure of Wold decomposition

The source's proof of Proposition 5.1 applies verbatim. Its lower bound for $\alpha_\lambda((0,m))$ diverges because every $p_m(j) \geq 1$; hence S is analytic. For the Cauchy-dual weights it computes

$$\alpha_{\lambda'}((0,0)) = \sum_{n=1}^{\infty} \frac{1}{p_{n+1}(n-1)} = 1 + \sum_{n=2}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty.$$

Theorem 1.3 of the source says that an analytic shift in this setting has Wold-type decomposition exactly when its Cauchy dual is analytic, equivalently when these $\alpha_{\lambda'}$ series diverge. Thus S has no Wold-type decomposition. This proves the theorem. $\square$

6 Scope and verification

The universal question is fully answered in the negative; no claim is made about the still sharper order-3 case. Exact rational code independently propagates the tree weights, checks norm increase and $\Delta_4 = 0$ on a large two-dimensional window, and verifies the displayed third-defect formula.

The earlier arXiv:2212.04446 claimed an order-3 counterexample, but its v3 metadata says that the paper was withdrawn because the main proof has a gap. Bounded exact-phrase and structural searches found no prior order-4 counterexample. Novelty is therefore plausible but provisional.

References

- [1] S. Chavan and S. Trivedi, *Wold-type decomposition for left-invertible weighted shifts on a rootless directed tree*, arXiv:2501.01296 (2025).
- [2] S. Chavan and S. Trivedi, *Failure of the wandering subspace property for analytic norm-increasing 3-isometries*, arXiv:2212.04446v3 (withdrawn).